

MH1301 Discrete Mathematics

Handout 3: Binomial Theorem, Generalized Permutations and Combination

This handout covers the remaining topics in Part I: Counting. These are topics extending from our earlier discussion on combinations and permutations.

- Binomial Theorem
- Division Rule
- Permutation around a Cycle
- Permutation and Combination with Repetition
- Permutations with indistinguishable objects

Binomial Coefficients

What is the expansion of $(x + y)^2$?

Solution:

$$\begin{aligned}(x + y)^2 &= (x + y)(x + y) = x \cdot x + x \cdot y + y \cdot x + y \cdot y \\ &= x^2 + 2xy + y^2.\end{aligned}$$

What is the expansion of $(x + y)^3$?

Solution:

$$\begin{aligned}(x + y)^3 &= (x + y)(x + y)(x + y) \\ &= x^3 + 3x^2y + 3xy^2 + y^3.\end{aligned}$$

What is the expansion of $(x + y)^n$?

Binomial Theorem

Theorem 1 (Section 6.4) The Binomial Theorem

Let x and y be variables and let n be a nonnegative integer. Then

$$\begin{aligned}(x + y)^n &= C(n, 0)x^n + C(n, 1)x^{n-1}y + \cdots + C(n, n-1)xy^{n-1} + C(n, n)y^n \\ &= \sum_{i=0}^n C(n, i)x^{n-i}y^i.\end{aligned}$$

$\binom{n}{n-k} x^{n-k} y^k$

Recall that $C(n, i)$ is also called the **binomial coefficient** and also denoted using $\binom{n}{i}$.

Binomial Theorem

Combinatorial Proof:

- We expand $\overbrace{(x + y)(x + y) \cdots (x + y)}^{n\text{-copies}}$ and count the number of terms of the form $x^{n-i}y^i$.
- To obtain such a term, it is necessary to choose $n - i$ many x 's from the n sums (so that the other i terms in the product are y 's).
- Therefore, the coefficient of $x^{n-i}y^i$ is $\underline{C(n, n - i)}$, which is equal to $C(n, i)$.



Example

Example 2 (Section 6.4)

What is the expansion of $(x + y)^4$?

Solution:

$$\begin{aligned}(x + y)^4 &= \sum_{i=0}^4 C(4, i)x^{4-i}y^i \\&= C(4, 0)x^4 + C(4, 1)x^3y + C(4, 2)x^2y^2 + C(4, 1)xy^3 + C(4, 0)y^4 \\&= x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4.\end{aligned}$$

Example

Example 3 (Section 6.4)

What is the coefficient of $x^{12}y^{13}$ in the expansion of $(x + y)^{25}$?

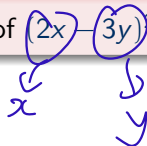
Solution: By the binomial theorem, the coefficient is

$$C(25, 13) = \frac{25!}{13!12!} = 5,200,300.$$

Example

Example 4 (Section 6.4)

What is the coefficient of $x^{12}y^{13}$ in the expansion of $(2x - 3y)^{25}$?



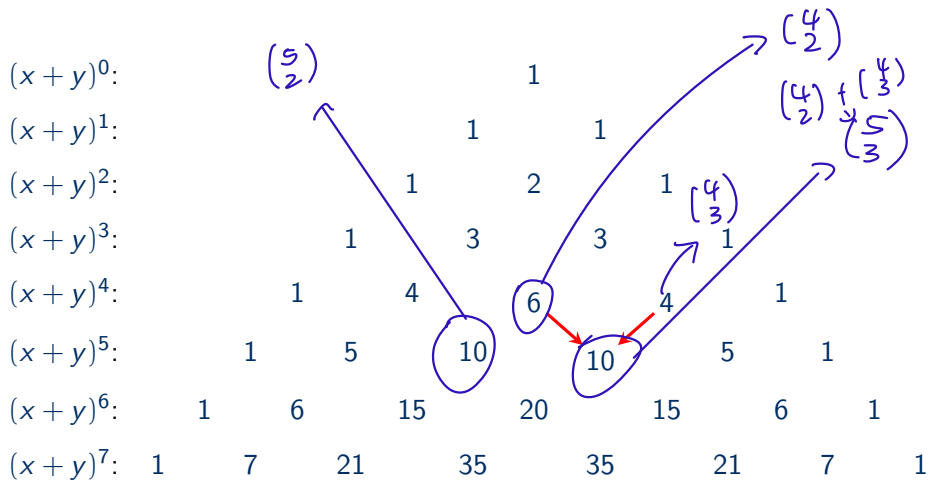
Solution: By the binomial theorem, we have

$$(2x - 3y)^{25} = [(2x) + (-3y)]^{25} = \sum_{i=0}^{25} C(25, i)(2x)^{25-i}(-3y)^i.$$

Hence, the coefficient of $x^{12}y^{13}$ is obtained when $i = 13$. That is

$$C(25, 13)2^{12}(-3)^{13} = -\frac{25!}{13!12!}2^{12}3^{13}.$$

Pascal's Triangle



watch again

Application

Corollary 1 (Section 6.4)

Let n be a nonnegative integer. Then

$$\sum_{i=0}^n C(n, i) = 2^n.$$

$$(x+y)^n = (1+1)^n = 2^n$$

Proof: Using the binomial theorem with $x = 1$ and $y = 1$, we have

$$\begin{aligned} 2^n &= (1 + 1)^n \\ &= \sum_{i=0}^n C(n, i) 1^{n-i} 1^i \\ &= \sum_{i=0}^n C(n, i). \end{aligned}$$



Application - Combinatorial Proof

Corollary 1 (Section 6.4)

Let n be a nonnegative integer. Then

$$\sum_{i=0}^n C(n, i) = 2^n.$$

Combinatorial Proof: Let S be a set with n elements.

- On the right hand side, 2^n is the total number of subsets of S .
- On the left hand side, there are
 - ▶ $C(n, 0)$ subsets with 0 elements,
 - ▶ $C(n, 1)$ subsets with 1 elements,
 - ▶ ...
 - ▶ $C(n, n)$ subsets with n elements.

So both sides are counting the same set of objects. The claim is proved. $\square$

Pascal's Identity

Theorem 2 (Section 6.4) Pascal's Identity

Let n and r be positive integers with $n \geq r$, then

$$C(n+1, r) = C(n, r) + C(n, r-1)$$

Proof:

$$\begin{aligned} C(n, r) + C(n, r-1) &= \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!} \\ &= \frac{n!}{(r-1)!(n-r)!} \left(\frac{1}{r} + \frac{1}{n-r+1} \right) \\ &= \frac{n!}{(r-1)!(n-r)!} \cdot \frac{n+1}{r(n-r+1)} \\ &= \frac{(n+1)!}{r!(n-r+1)!} = C(n+1, r). \end{aligned}$$



Pascal's Identity - Combinatorial Proof

$$C(n+1, r) = C(n, r) + C(n, r-1)$$

Combinatorial Proof:

- Assume that S is a set with $n+1$ elements, then $C(n+1, r)$ is the number of subsets of S with r elements.
- Pick a specific element $a \in S$, and let $T = S - \{a\}$.
- Consider again a subset of S with r elements
 - ▶ either the subset contains a together with $r-1$ elements of T , in which there are $C(n, r-1)$ such possibilities.
 - ▶ or the subset contains r elements of T and does not contain a , in which there are $C(n, r)$ such possibilities.
- Consequently, $C(n+1, r) = C(n, r) + C(n, r-1)$.



Generalized Permutations and Combinations

Practical counting problems may be more complex.

- elements may be placed in a cycle,
- elements may be used repeatedly,
- may involve indistinguishable elements,
- etc.

How to count these sets?

- view the problem from different perspectives,
- apply/combine different counting rules,
 -  Do not overcount or undercount!
- use bijective functions to reduce the problem to known cases.

Division Rule

A function $f : X \rightarrow Y$ is *k-to-1* if for every $y \in Y$, the number of $x \in X$ such that $f(x) = y$ is exactly k .

The Division Rule (Section 6.1)

If function $f : X \rightarrow Y$ is *k-to-1*, then $|Y| = \frac{|X|}{k}$.

If we want to count the size of Y and have a *k-to-1* function from X to Y , where X is a set whose size we know, we can conclude that

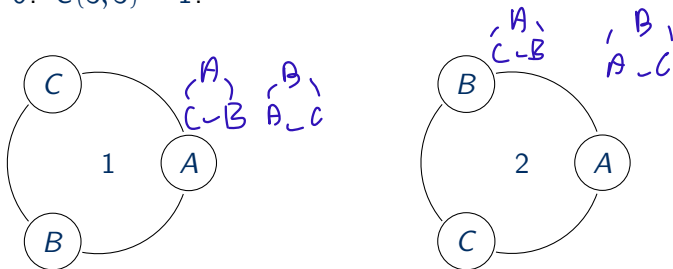
$$|Y| = \frac{|X|}{k}.$$

Example - Table

Question

How many ways are there to arrange 3 students to sit around a table?

$$P(3, 3) = 6? \quad C(3, 3) = 1?$$



Question

How do we select r students from a set of n students and arrange them around a table?

Permutation around a Cycle

Theorem (Exercise 6.3.31)

The number of r -permutations of a set with n elements around a cycle is

$$\frac{1}{r} \cdot P(n, r) = \frac{n!}{r(n-r)!}.$$

Proof:

- Let P be the set of all r -permutations with n elements and C be the set of all r -permutations with these elements around a cycle.
- Define $f : P \mapsto C$, such that for each r -permutation $(p_1, \dots, p_r)$, $f(p_1, \dots, p_r)$ is the circular permutation obtained by placing these elements around a cycle in the same order.
 - ▶ e.g. when $n = 3$ and $r = 3$, let $S = \{A, B, C\}$,
 f maps $(A, B, C), (B, C, A), (C, A, B)$ to circular permutation 1,
and $(A, C, B), (C, B, A), (B, A, C)$ to circular permutation 2.



Permutation around a Cycle

Proof: (continue)

- Function f is r -to-1. This is because the following r permutations map to the same circular permutation

$$\begin{array}{cccccccc} a_1 & a_2 & \dots & \dots & \dots & a_r & & \\ & a_2 & a_3 & \dots & \dots & a_r & a_1 & \\ & & a_3 & a_4 & \dots & a_r & a_1 & a_2 \\ & & & & \dots & \dots & & \\ & & & & & a_r & a_1 & a_2 & \dots & a_{r-1} \end{array}$$

- Since $|P| = P(n, r)$, by the division rule we know that

$$|C| = \frac{P(n, r)}{r} = \frac{n!}{r(n-r)!}.$$



Example - Table

How many ways are there to arrange 3 students to sit around a table?

Solution: Every assignment is a 3-permutation around a cycle from a set with 3 elements.

Thus the answer is

$$\frac{P(3, 3)}{3} = 2.$$

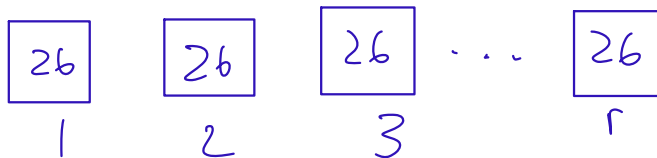
Example - English Words

Example 1 (Section 6.5)

How many strings of length r can be formed from the English alphabet?

Solution: In each position of the word, there are 26 letters for each position.

By the product rule, there are in total 26^r strings of length r .



Permutations with Repetition

Theorem 1 (Section 6.5)

The number of r -permutations of a set of n elements with repetition allowed is n^r .

Proof: For each of the r positions, there are n ways to select an element from the set (because repetition is allowed).

By the product rule, there are n^r r -permutations when repetition is allowed. □

Example - Fruits

How many ways are there to select 3 pieces of fruit from a bowl containing apples, oranges and pears if the order in which the pieces are selected does not matter?

Solution: By counting manually, we find 10 different ways:

Apple	Orange	Pear
3	0	0
0	3	0
0	0	3
2	1	0
2	0	1

Apple	Orange	Pear
1	2	0
0	2	1
1	0	2
0	1	2
1	1	1

Combinations with Repetition

Question

How do we count the number of r -combinations with repetition allowed?

View it from a different perspective.

- Assume we have 1 bin for each element from set S , so n bins in total.

select 2 apples and 1 orange



the “apple” bin has 2 balls and the “orange” bin has 1 ball



Combinations with Repetition

count ways to pick r -combinations with repetition allowed



count ways to fill these n bins with r balls



place r balls together with $n - 1$ dividers

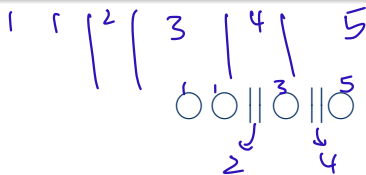


Combinations with Repetition

Question

Let $S = \{1, 2, 3, 4, 5\}$, how do we represent the 4-combination with repetition $\{1, 1, 3, 5\}$?

Solution:



An r -combination with repetition allowed is equivalent to a placement of the r balls ($\bigcirc$) and $n - 1$ dividers ($|$).

The $n - 1$ dividers are used to mark off n different bins, where balls in the i th bins corresponds to the number of element i selected in the r -combination.

Combinations with Repetition

Question

Let $S = \{1, 2, 3, 4, 5\}$, what combination does “ $\bigcirc || \bigcirc | \bigcirc \bigcirc |$ ” represent?

Solution: It represents the 4-combination with repetition $\{1, 3, 4, 4\}$.

Question

How many sequences are there with $n + r - 1$ elements that contain r balls and $n - 1$ dividers?

*↓ can be less than n
cos we allow repetition*

Answer is in the next theorem.

Combinations with Repetition

Theorem 2 (Section 6.5)

There are $C(n+r-1, r) = C(n+r-1, n-1)$ r -combinations from a set with n elements when repetition is allowed.

Proof:

$\downarrow r + (n-1)$
what to choose $\rightarrow$ dividers

- Every r -combination of a set with n elements when repetition is allowed can be viewed as a sequence of $n+r-1$ elements that contain r balls and $n-1$ dividers.
- Every sequence of $n+r-1$ elements that contain r balls and $n-1$ dividers also corresponds to an r -combination with repetition.
- We have a bijective function.
- Number of sequence of $n+r-1$ elements that contain r balls and $n-1$ dividers is

$$C(n+r-1, r) = C(n+r-1, n-1).$$

Example - Fruits

How many ways are there to select 3 pieces of fruit from a bowl containing apples, oranges and pears if the order in which the pieces are selected does not matter?

Solution: We are asked to count the number of 3-combinations of a set of 3 elements with repetition, so the answer is

$$C(3 + 3 - 1, 3) = C(5, 3) = 10. = C(5, 2)$$

Example - Cookies

Example 4 (Section 6.5)

A cookie shop has 4 different kinds of cookies. How many ways can 6 cookies be chosen if the order in which they are chosen does not matter?

Solution: We are asked to count the number of 6-combinations of a set of 4 elements with repetition, so the answer is

$$C(4 + 6 - 1, 6) = C(9, 6) = 84. = C(9, 3)$$

Example - Equations

Example 5 (Section 6.5)

How many solutions does the equation $x_1 + x_2 + x_3 = 11$ have, where x_1, x_2, x_3 are nonnegative integers?

$$n+r-1 = 3+11-1 = 13$$

$$\hookrightarrow \binom{13}{11}$$

$$n=3$$

$$r=11$$

Solution:

- A solution corresponds to a way of selecting 11 items from a set with 3 types of elements, i.e., x_1 items of type one, x_2 items of type two and x_3 items of type three.
- Hence, the number of solutions is equal to the number of 11-combinations with repetition from a set of 3 elements, and the answer is

$$C(3 + 11 - 1, 11) = C(13, 11) = C(13, 2) = 78.$$

Example - Equations

How many solutions does the equation $x_1 + x_2 + x_3 = 11$ have, where $x_1 \geq 1$, $x_2 \geq 2$ and $x_3 \geq 3$ are integers?

Solution:

- Define $x_1 = 1 + y_1$, $x_2 = 2 + y_2$, $x_3 = 3 + y_3$.
(Handwritten: $y_1 \geq 0$, $y_2 \geq 0$, $y_3 \geq 0$)
- The equation is equivalent to $y_1 + y_2 + y_3 = 5$, where y_1, y_2, y_3 are nonnegative integers.
(Handwritten: $\downarrow n=3$ with an arrow pointing to the 5)
- Hence, it's the number of 5-combinations with repetition from a set of 3 elements, and the answer is

You have a bag containing red, blue, and green marbles. If the order in which the marbles are selected doesn't matter, how many ways can you choose 5 marbles from the bag?

Answer = 21
(Handwritten: $\Rightarrow n=3$, $r=5$)
$$\binom{3+5-1}{5} = \binom{7}{5} = \binom{7}{2} = 21$$

How many solutions does the equation $x+y+z=6$ have, where x, y , and z are nonnegative integers (i.e. at least 0)?

Answer = 28
(Handwritten: $n=3, r=6$)
$$\binom{3+6-1}{6} = \binom{8}{6} = \binom{8}{2} = 28$$

$$C(3 + 5 - 1, 5) = C(7, 5) = C(7, 2) = 21.$$

(Handwritten: "cashe type wrong in i think")
$$n=3, r=3 \quad \left| \quad n=3, r=3 \right. \\ \binom{3}{2} = 3 \quad \left| \quad \binom{5}{2} = 10 \rightarrow \text{accepted also??}$$

Example - SUCCESS

Example 7 (Section 6.5)

How many strings can be made by reordering the letters of the word "SUCCESS"?



A handwritten calculation in blue ink. It shows $7!$ over a horizontal line, with $2! \times 3!$ below the line. An arrow points from the first 'S' in the word 'SUCCESS' in the text above to the $7!$ in the calculation.

$$\frac{7!}{2! \times 3!}$$

Observation:

- There are 2 "C"s and 3 "S"s.
- It is NOT a permutation with 7 letters or 4 letters!
 - ▶ why not?

Example 7 (Section 6.5)

How many strings can be made by reordering the letters of the word "SUCCESS"?

Solution: In any string, there are 7 positions - three "S"s, two "C"s, one "U", and one "E".

- Three "S"s can be placed among the seven positions in $C(7, 3)$ different ways, leaving four positions free.
- Then the two "C"s can be placed in $C(4, 2)$ ways.
- Then "U" can be placed in $C(2, 1)$ ways.
- Finally, "E" can be placed in $C(1, 1)$ way.

Consequently, by the product rule, the final answer is

$$\begin{aligned} C(7, 3)C(4, 2)C(2, 1)C(1, 1) &= \frac{7!}{3!4!} \cdot \frac{4!}{2!2!} \cdot \frac{2!}{1!1!} \cdot \frac{1!}{1!0!} \\ &= \frac{7!}{3!2!1!1!} = 420. \end{aligned}$$

Permutation with Indistinguishable Objects

Theorem 3 (Section 6.5)

The number of different permutations of n objects, where there are

- n_1 indistinguishable objects of type 1,
- n_2 indistinguishable objects of type 2,
- ...
- n_k indistinguishable objects of type k ,

is

$$\frac{n!}{n_1!n_2!\cdots n_k!}.$$

Proof of Theorem 3:

- Let C be the set of permutations of these n objects.
- Let P be the set of permutations of these n objects when we view them as distinguishable objects.
 - e.g., view the three letter "S"s as S_1, S_2, S_3
 - $|P| = n!$
- Define $f : P \rightarrow C$ such that every permutation with distinguishable objects is mapped to the corresponding permutation with indistinguishable objects
 - e.g. $f(S_1UC_1C_2ES_2S_3) = f(S_1UC_2C_1ES_3S_2) = SUCCESS$
- How many elements in P are mapped to the same element in C ?
 - the relative orders of S_1, S_2, S_3 and C_1, C_2 do not matter
- Function f is $n_1!n_2! \cdots n_k!$ -to-1.
- By the division rule, $|C| = \frac{|P|}{n_1!n_2! \cdots n_k!} = \frac{n!}{n_1!n_2! \cdots n_k!}$.

$U S S S C C E$
 $U S S S C C E$
 $U S S S C C E$



Example - MISSISSIPPI

Ex.6.5.24

How many strings can be made by reordering the letters of the word "MISSISSIPPI"?

11 letters

Solution: A valid string is a permutation with

- 4 elements of type "I"
- 4 elements of type "S"
- 2 elements of type "P"
- 1 element of type "M".

So the number of different strings is

$$\frac{11!}{4! \times 4! \times 2!} = 34,650.$$

Distributing Distinguishable Objects into Distinguishable Boxes

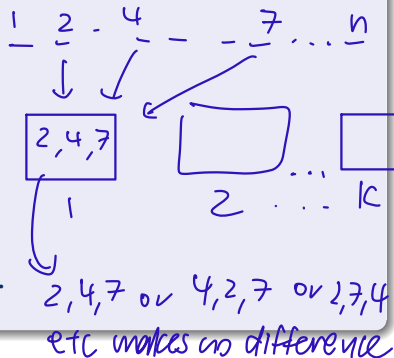
Theorem 4 (Section 6.5)

The number of ways to **distribute** n distinguishable objects into k distinguishable boxes so that

- n_1 objects are placed in box 1,
- n_2 objects are placed in box 2,
- ...
- n_k objects are placed in k ,

is

$$\frac{n!}{n_1! n_2! \cdots n_k!}$$



See Ex 6.5.36 for a proof (by setting up a 1-1 bijection between the permutations counted in Theorem 3 and those counted in Theorem 4).

Distributing Indistinguishable Objects into Distinguishable Boxes

Example

How many ways are there to distribute 5 indistinguishable objects into 3 distinguishable boxes?

Solution: The answer turns out to be the number of 5-combinations for a set with 3 elements with repetition allowed, which is

$$C(3 + 5 - 1, 5) = C(7, 5) = C(7, 2) = 21.$$

↓ ↓
n r

Summary

	r -permutation	r -combination
w/o repetition	$P(n, r) = \frac{n!}{(n-r)!}$	$C(n, r) = \frac{n!}{(n-r)!r!}$
w/ repetition	n^r Thm 1 slide 21	$C(n+r-1, r)$ Thm 2 slide 27
w/ indistinguishable objects	$\frac{n!}{n_1!n_2!\cdots n_k!}$ Thm 3 slide 34	also Thm 4 n dist. Obj into k dist boxes
around a cycle	$\frac{P(n, r)}{r} = \frac{n!}{r(n-r)!}$	← slide 17

Brief Review of Counting

- Counting Rules

- ▶ sum rule
- ▶ product rule
- ▶ division rule
- ▶ bijective function mapping rule

- Inclusion-exclusion principle

- Binomial theorem

- Combinatorial proof

- Permutations/Combinations

- Generalizations

- ▶ permutations around a cycle
- ▶ permutations/combinations with repetitions

Be careful with the conditions associated with each counting rule.

Do not undercount or overcount.

The screenshot shows a mobile app interface with a dark status bar at the top displaying '16:52', 'app.moorcap.com', and '4G'. The main content area is titled 'Answer these questions.' and includes a 'Waiting for next clap' indicator. There are three questions listed, each with a handwritten solution in blue ink:

- Question 1: 'Number of ways to reorder the letters of the word "TOMORROW" = 3360'. Handwritten solution: $\frac{8!}{3! \times 2!}$.
- Question 2: 'Number of ways to distribute 8 indistinguishable objects into 4 distinguishable boxes = 165'. Handwritten solution: $C(4+8-1, 8)$.
- Question 3: 'Find the number of ways to distribute 8 distinguishable objects into 4 distinguishable boxes so that there are 2 objects in each box. = 2520'. Handwritten solution: $8! / (2! \times 2! \times 2! \times 2!)$.

At the bottom of the app interface, there is a link that says 'Try Woodclap for free'.

Reading Assignment

Rosen (7th edition):

- Chapter 6.4, Binomial Coefficients and Identities
- Chapter 6.5, Generalized Permutations and Combinations